# CSA 0378-DATA STRUCTURES

# 10. STACK USING ARRAY

main.c

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value)
{
    if(top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }

    stack[++top] = value;
    printf("%d pushed into stack\n", value);
}

void pop()
{
    if(top == -1)
    {
        printf("Stack Underflow\n");
        return;
    }

    printf("Popped Element: %d\n", stack[top--]);
}

void display()
{
    int i;

    if(top == -1)
    {
```

Output

```
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice:
```

main.c


```c
void display()
{
    int i;

    if(top == -1)
    {
```


```c
        printf("Stack is Empty\n");
        return;
    }

    printf("Stack Elements:\n");

    for(i = top; i >= 0; i--)
        printf("%d ", stack[i]);

    printf("\n");
}

int main()
{
    int choice, value;

    while(1)
    {
        printf("\n1. Push\n");
        printf("2. Pop\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter Choice: ");
        scanf("%d", &choice);

        if(choice == 1)
        {
            printf("Enter Value: ");
            scanf("%d", &value);
            push(value);
```

Output


```
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice:
```


main.c


```c
    while(1)
    {
        printf("\n1. Push\n");
        printf("2. Pop\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter Choice: ");
        scanf("%d", &choice);

        if(choice == 1)
        {
            printf("Enter Value: ");
            scanf("%d", &value);
            push(value);
```


```c
        }
        else if(choice == 2)
        {
            pop();
        }
        else if(choice == 3)
        {
            display();
        }
        else if(choice == 4)
        {
            break;
        }
        else
        {
            printf("Invalid Choice\n");
        }
    }

    return 0;
}
```

Output


```
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice:
```